

Zadanie 22

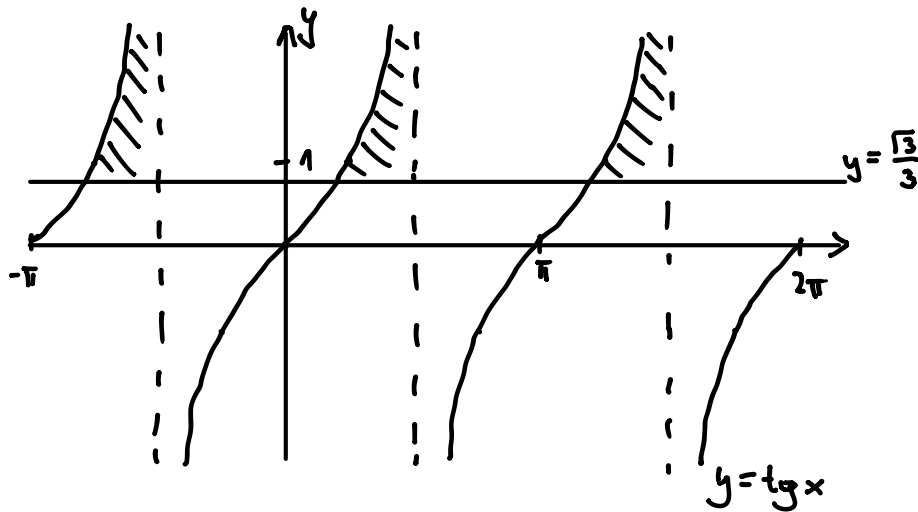
Rozwiąż nierówność $\operatorname{tg} x \geq \operatorname{tg} \frac{7\pi}{6}$, gdy $x \in [-\pi, 2\pi]$.

① Obliczamy $\operatorname{tg} \frac{7\pi}{6}$.

$$\operatorname{tg} \frac{7\pi}{6} = \operatorname{tg}(\pi + \frac{\pi}{6}) = \operatorname{tg} \frac{\pi}{6} = \frac{\sqrt{3}}{3}$$

$$\operatorname{tg} x \geq \frac{\sqrt{3}}{3}$$

② Szkicujemy wykres $\operatorname{tg} x$.



$$x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$$

③ Odczytujemy rozwiązanie z wykresu.

$$x \in [-\frac{5\pi}{6}, -\frac{\pi}{2}) \cup [\frac{\pi}{6}, \frac{\pi}{2}) \cup [\frac{7\pi}{6}, \frac{3\pi}{2})$$

$$\text{Odp: } x \in [-\frac{5\pi}{6}, -\frac{\pi}{2}) \cup [\frac{\pi}{6}, \frac{\pi}{2}) \cup [\frac{7\pi}{6}, \frac{3\pi}{2}).$$